Akoizo Original Practice - Form 2

Division B | Solar System | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) A planet moves from 1 AU to 2 AU. By what factor does the starlight flux change?

A. It doubles

B. It becomes 1/8 as large

C. It becomes 1/4 as large

D. It becomes 1/2 as large

Answer: _____

2. (2 points) Which moon has its own internally generated magnetic field?

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

A. Titan

B. Enceladus

C. Ganymede

D. Europa

Answer: _____

3. (2 points) Why are small, hot worlds poor at keeping light gases?

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

A. Gas molecules move fast compared with a low escape velocity

B. Gravity is too strong

C. Light gases freeze out

D. The Sun adds gas

Answer: _____

4. (2 points) A high D/H ratio in Venus's atmosphere suggests:

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

A. Venus lost a lot of water over time

B. Venus has oceans today

C. Comets just delivered water

D. The atmosphere is mostly hydrogen

Answer: _____

5. (2 points) Which telescope used transmission spectroscopy to study TRAPPIST-1 planets?

A. JWST

B. Kepler

C. Rosetta

D. MAVEN

Answer: _____

6. (2 points) Tidal force depends on distance as:

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

A. 1/d^2

B. d^2

C. 1/d^3

D. 1/d

Answer: _____

7. (2 points) An organism at a dark hydrothermal vent that uses hydrogen sulfide for energy is a:

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

A. Tardigrade

B. Heterotroph that eats plants

C. Chemolithoautotroph

D. Photoautotroph

Answer: _____

8. (4 points) Why is Earth's actual average temperature higher than its equilibrium temperature?

Answer: _____
_____
_____

9. (4 points) Why is Europa considered a strong candidate for habitability?

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

Answer: _____
_____
_____

10. (4 points) Describe how orbital resonance keeps tidal heating going.

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

Answer: _____
_____
_____

11. (4 points) How did losing its magnetic field affect Mars?

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

Answer: _____
_____
_____

12. (4 points) Why can microlensing not be used to follow up a planet?

Unit: Exoplanets: detection and the named systems.
Worked-case data: An Earth-sized planet transits two stars: a Sun-like star and a red dwarf with one-tenth the Sun's radius. Compare the transit depths.
Separate investigation observations:
Light curve: Seven different periodic dips in brightness.
Star: An ultracool red dwarf, about 40 light-years away.
Sizes: All planets are roughly Earth-sized.
JWST: The two innermost planets show no thick atmosphere.

Answer: _____
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13. (4 points) Why do planets in a red dwarf's habitable zone have short orbital periods?

Answer: _____
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14. (4 points) Explain why extremophiles matter for habitability studies.

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
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Isotopes: D/H ratio close to Earth's oceans.

Answer: _____
_____
_____

15. (4 points) A spectral line appears at a shorter wavelength than expected. What does this mean?

Unit: Light, radiation, and temperature.
Worked-case data: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth's 255 K.
Separate investigation observations:
Distance: A planet orbits a Sun-like star at 0.72 AU.
Albedo: Very high: about 0.75, from bright clouds.
Predicted Teq: About 230 K, below freezing.
Measured surface: About 735 K.

Answer: _____
_____
_____

16. (4 points) What is unique about JUICE's planned orbit?

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

Answer: _____
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_____

17. (6 points) Solve this case and explain your reasoning: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.

Answer: _____
_____
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18. (6 points) Solve this case and explain your reasoning: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?

Answer: _____
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